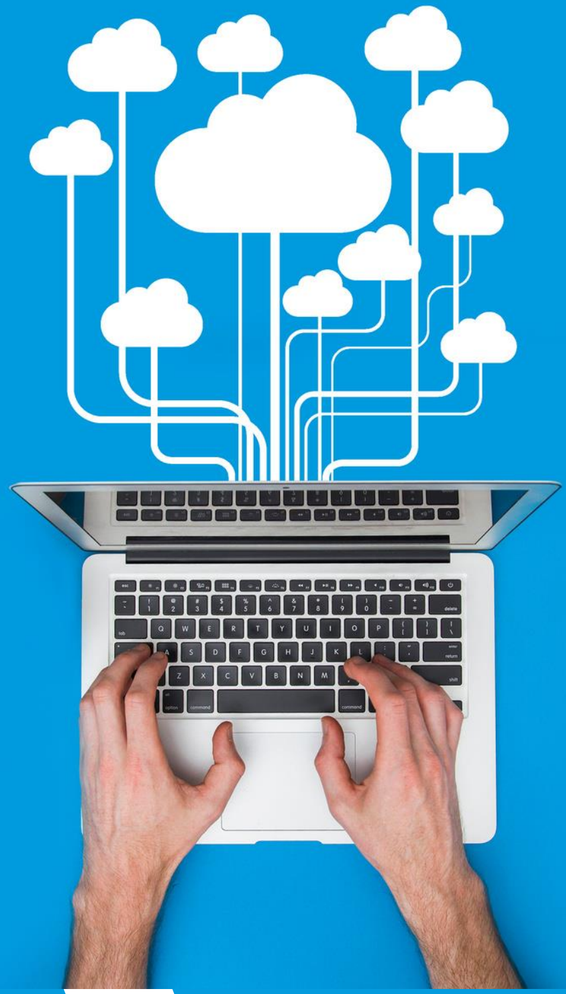
PYTHON PROGRAMMING



21 Days Challenge

Day-03



Day-03

PYTHON OPERATORS AND CHAINED COMPARISONS WITH LOGICAL OPERATORS

Instructor



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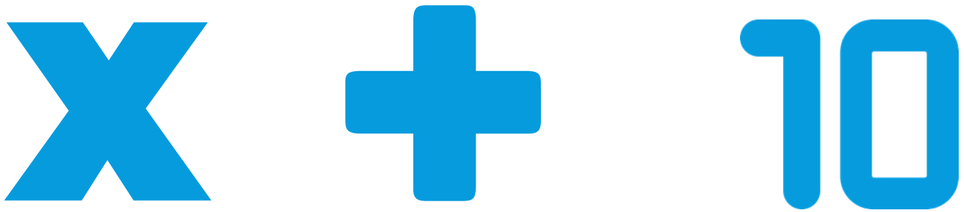
Table Of Content

- Operators
- Chaining Comparison
- Logic Control
- Operator Precedence

Python Operators

Operators are used to perform operations on variables and values.





Python Operators

- Arithmetic Operators
- Comparison Operators
- Assignment Operators
- Bitwise Operators
- Logical Operators
- Membership Operators
- Identity Operators

Arithmetic Operators

Arithmetic operators are used with numeric values to perform common mathematical operations.

a = 3, b = 2;

Operator	Name	Code Example	Output
+	Addition	print(a + b)	5
-	Subtraction	print(a - b)	1
*	Multiplication	print(a * b)	6
/	Division	print(a / b)	1.5
%	Modulus (Remainder)	print(a % b)	1
**	Exponentiation	print(a ** b)	9
//	Floor Division	print(a // b)	1 (1.5)

Comparison Operators

Comparison operators are used to compare two values

a = 2, b = 3;

Operator	Name	Code Example	Output
=	Equal	print(a == b)	FALSE
!=	Not equal	print(a != b)	TRUE
>	Greater than	print(a > b)	FALSE
<	Less than	print(a < b)	TRUE
>=	Greater than or equal to	print(a >= b)	FALSE
<=	Less than or equal to	print(a <= b)	TRUE

Logical Operators

Logical operators are used to combine conditional statements

a = 10, b = 20 ;

Operator	Name	Code Example	Output
and	AND	(a > 5) and (b > 15)	TRUE
or	OR	(a > 15) or (b > 15)	TRUE
not	NOT	not(a > 15)	TRUE

Membership Operators

Membership operators are used to test if a sequence is presented in an object.

a = 5

b = [1, 2, 3, 4, 5]

Operator	Name	Code Example	Output
in	Membership IN	a in b	TRUE
not in	Membership NOT IN	a not in b	FALSE

Identity Operators

Returns True if two variables point to the same object in memory.

a = [1, 2, 3]
b = a
c = [1, 2, 3]

Operator	Name	Code Example	Output
is	Identity IS	a is b	TRUE
is	Identity IS	a is c	FALSE
is not	Identity IS NOT	a is not c	TRUE
is not	Identity IS NOT	a is not b	FALSE

Bitwise Operators

Bitwise operators are used to compare (binary) numbers.

a = 10 # bin(a) → '0b1010' , b = 4 # bin(b) → '0b100'

Operator	Name	Code	Output
&	Bitwise AND	print(a & b)	0(0b0000)
|	Bitwise OR	print(a | b)	14(0b1110)
^	Bitwise XOR	print(a ^ b)	14(0b1110)
~	Bitwise NOT	print(~a)	-11(-0b1011)
<<	Left Shift	print(a << 2)	40(0b101000)
>>	Right Shift	print(a >> 2)	2(0b10)

1 0 1 0
0 1 0 0

Assignment Operators

Assignment operators are used to assign values to variables

Operator	Name	Example	Same As	Output (if a = 10)
	Assignment	a = 10	a = 10	10
+=	Add and Assign	a += 5	a = a + 5	15
-=	Subtract and Assign	a -= 3	a = a - 3	7
*=	Multiply and Assign	a *= 2	a = a * 2	20
/=	Divide and Assign	a /= 2	a = a / 2	5
%=	Modulus and Assign	a %= 3	a = a % 3	1
**=	Power and Assign	a **= 2	a = a ** 2	100
//=	Floor Divide and Assign	a //= 3	a = a // 3	3
&=	Bitwise AND and Assign	a &= 7	a = a & 7	2
|=	Bitwise OR and Assign	a |= 5	a = a | 5	15
^=	Bitwise XOR and Assign	a ^= 5	a = a ^ 5	15
>>=	Right Shift and Assign	a >>= 1	a = a >> 1	5
<<=	Left Shift and Assign	a <<= 1	a = a << 1	20

Chaining Comparison Operators

Chained Comparison (with/ without Logical Operators)

Expression	**Result**
5 < x < 15	TRUE
x > 5 and x < 15	TRUE
x < 5 or x > 15	FALSE
not (x > 5 and x < 15)	FALSE

Operators Precedence

Determine the priorities of the operator.

Precedence: () > ** > *, /, //, % > +, - >
comparison operators (==, !=, <, >, <=, >=) >
not > and > or > assignment (=, +=, -=, etc.)

Example: x = 5 + 2 * 3 ** 2 // 4 – 1

3 ** 2 → 9
2 * 9 → 18
18 // 4 → 4 (floor division)
5 + 4 → 9
9 - 1 → 8
x = 8

QUESTION SESSION

THANK YOU

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